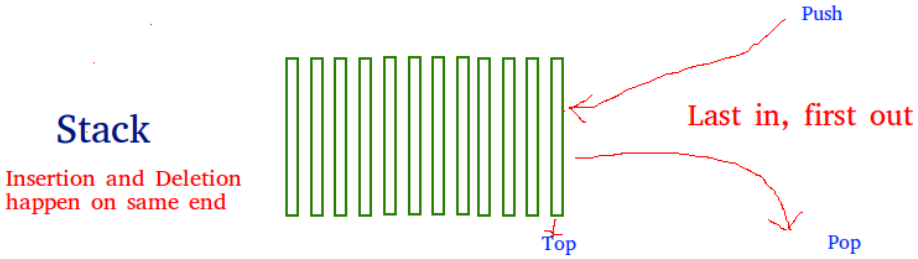
Lab 9: Stack



Part A: Implement Stack using Array and Linked-List

1. Assume below is a series of stack operations in a C++ program.

push('TP01'); push('TP02'); push('TP03'); pop(); pop(); push('TP04'); push('TP05'); push('TP03'); pop(); push('TP02');

a) Draw a step-by-step diagram to describe the below operations and evaluate the final answer in the stack after the operations are executed.

b) According to the above specific steps, write a program using linked list-based stack implementation to verify the answer in (a).

c) According to the above specific steps, write a program using array-based stack implementation to verify the answer in (a).

Example Output:

```
Push TP01 :
Current List: TP01 ,

Push TP02 :
Current List: TP01 , TP02 ,

Push TP03 :
Current List: TP01 , TP02 , TP03 ,

Pop TP03: TP03 is deleted!
Current List: TP01 , TP02 ,

Pop TP02: TP02 is deleted!
Current List: TP01 ,

Push TP04 :
Current List: TP01 , TP04 ,

Push TP05 :
Current List: TP01 , TP04 , TP05 ,

Push TP03 :
Current List: TP01 , TP04 , TP05 , TP03 ,

Pop TP03: TP03 is deleted!
Current List: TP01 , TP04 , TP05 ,

Push TP02 :
Current List: TP01 , TP04 , TP05 , TP02 ,

From base to Top:
Current List: TP01 , TP04 , TP05 , TP02 ,

From Top to base :
Current List: TP02 , TP05 , TP04 , TP01 ,
```

[Estimate Finish Time: 60 minutes]

2. Write a program that reads in a sequence of characters, and determines whether its parentheses, braces, and curly braces are "balanced." Hint: for left delimiters, push onto stack; for right delimiters, pop from stack and check whether the popped element matches the right delimiter.

   Implements using Linked-List.

```
Microsoft Visual Studio Debug Console
Enter your equation here: ((a+b))+(

The parenthesis of the equation is not balance. Thus equation is not correct in term of writing.
```

```
Microsoft Visual Studio Debug Console
Enter your equation here: ((a+b))+((c+b))

The parenthesis of the equation is balance. Thus equation correct in term of writing.
```

[Estimate Finish Time: 60 minutes]

Part B: *Test Yourself Question*

Due date submission: _____

1. Write a program that reads in a sequence of characters and prints them in reverse order. **Use stack, implemented using linked lists and arrays respectively.**

2. Write a program that reads in a sequence of characters, and determines whether its parentheses, braces, and curly braces are "balanced." Hint: for left delimiters, push onto stack; for right delimiters, pop from stack and check whether the popped element matches the right delimiter. **Implements using Array.**

3. Given an infix expression in the form of string str. Convert this infix expression to postfix expression.

   - **Infix expression:** The expression of the form a **op** b. When an operator is in-between every pair of operands.
   - **Postfix expression:** The expression of the form a b **op**. When an operator is followed for every pair of operands.

     Note: The order of precedence is: ^ **greater than** * **equals to** / **greater than** + **equals to** -. Ignore the right associativity of ^.

Example 1:

Input: str = "a+b*(c^d-e)^(f+g*h)-i"
Output: abcd^e-fgh*+^*+i-
Explanation:
After converting the infix expression into postfix expression, the resultant expression will be abcd^e-fgh*+^*+i-

Example 2:

Input: str = "A*(B+C)/D"
Output: ABC+*D/
Explanation:
After converting the infix expression into postfix expression, the resultant expression will be ABC+*D/

```
Microsoft Visual Studio Debug Console
Enter your equation here: (A+B)*(C+D)

The postfix of the equation = AB+CD+*

C:\Users\mienmay\source\repos\Project12\Debug\Project12.exe (process 19964) exited with code 0.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close t
he console when debugging stops.
Press any key to close this window . . .
```

```
Microsoft Visual Studio Debug Console
Enter your equation here: A*B+C*D

The postfix of the equation = AB*CD*+

C:\Users\mienmay\source\repos\Project12\Debug\Project12.exe (process 103
48) exited with code 0.
To automatically close the console when debugging stops, enable Tools->O
ptions->Debugging->Automatically close the console when debugging stops.
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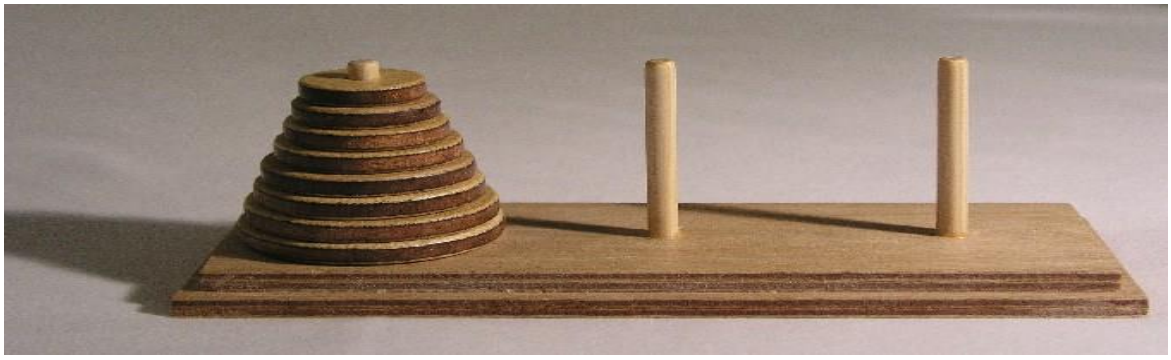
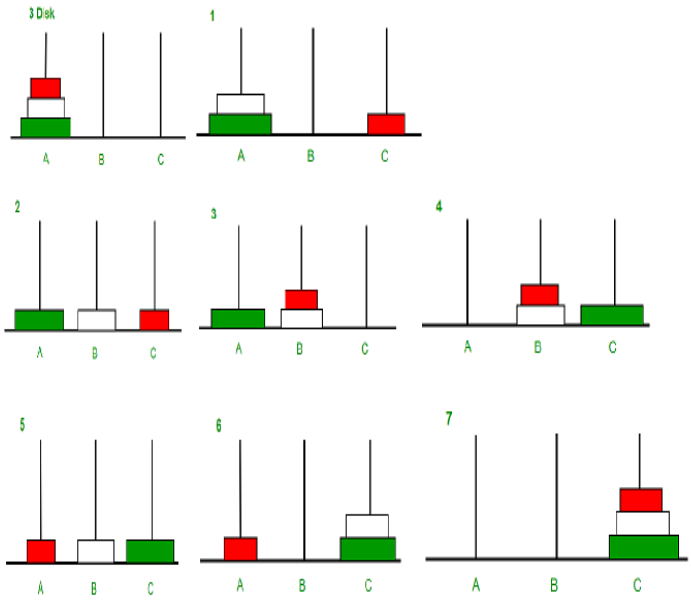
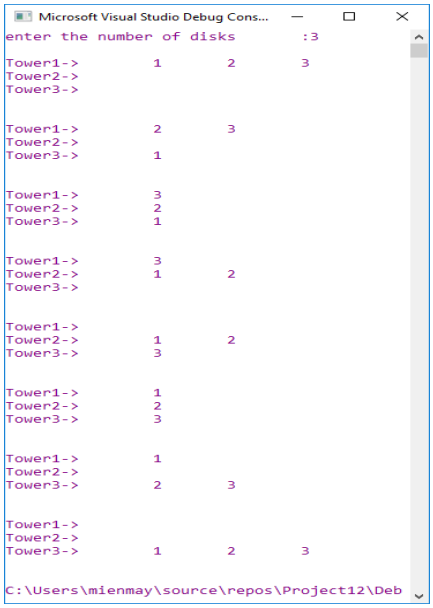
4. The **Tower of Hanoi** (also called the **Tower of Brahma** or **Lucas' Tower**, and sometimes pluralized) is a mathematical game or puzzle.



The objective of the puzzle is to move the entire stack to another rod, obeying the following simple rules:

- Only one disk can be moved at a time.
- Each move consists of taking the upper disk from one of the stacks and placing it on top of another stack i.e. a disk can only be moved if it is the uppermost disk on a stack.
- No disk may be placed on top of a smaller disk.

Sample of movement: